Sanskrit-to-English Neural Machine Translation using Sequence-to-Sequence Architecture with Bahdanau Attention

Natural Language Understanding Assignment

Course: Natural Language Understanding

Assignment: Sanskrit-to-English Neural Machine Translation

Abstract

Neural Machine Translation (NMT) has become one of the most successful approaches for automatic language translation. This project implements a custom Sequence-to-Sequence (Seq2Seq) neural network with Bahdanau Attention for Sanskrit-to-English translation using the provided parallel corpus. The model was implemented in PyTorch and trained on 10,000 sentence pairs. Performance was evaluated using BLEU score, BERTScore, inference time, and model complexity. Although the model successfully learned the training data, the evaluation results indicate that Sanskrit-to-English translation remains challenging due to the limited dataset size and the rich morphology of Sanskrit.

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1. Introduction

Machine Translation is a Natural Language Processing (NLP) task that automatically converts text from one language into another while preserving meaning. Modern translation systems rely on Neural Machine Translation (NMT), which uses deep neural networks instead of statistical methods.

Sanskrit is a morphologically rich language with complex grammatical rules, making automatic translation significantly more difficult than for high-resource languages. The objective of this assignment was to develop a custom neural machine translation system capable of translating Sanskrit sentences into English using only the provided bilingual dataset.

The implemented system follows the Encoder-Decoder architecture with an attention mechanism, enabling the decoder to focus on important source words while generating the target sentence.

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2. Dataset Description

The dataset consists of parallel Sanskrit-English sentence pairs provided for the assignment.

Dataset	Number of Sentence Pairs
Training	10,000
Validation	1,000
Testing	1,000

Each sentence pair is linked using a common **Source_id**, ensuring correct alignment between Sanskrit and English sentences.

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3. Data Preprocessing

The following preprocessing steps were performed before training:

- Loading Sanskrit and English CSV files.
- Merging datasets using the common Source_id.
- Cleaning extra spaces and quotation marks.
- Tokenizing sentences using whitespace tokenization.
- Building separate vocabularies for Sanskrit and English.
- Adding special tokens:
    - <PAD>
    - <SOS>
    - <EOS>
    - <UNK>
- Converting words into numerical indices.
- Padding sequences to a maximum length of 50 tokens.

Vocabulary statistics:

- Sanskrit Vocabulary Size: **32,275**
- English Vocabulary Size: **16,661**

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4. Model Architecture

The proposed Neural Machine Translation model consists of four major components.

4.1 Encoder

The encoder uses a two-layer Bidirectional Long Short-Term Memory (Bi-LSTM) network.

The encoder:

- Converts Sanskrit tokens into embeddings.
- Processes the sentence in both forward and backward directions.
- Produces contextual hidden representations.

Embedding Dimension: **256**

Hidden Dimension: **512**

Number of Encoder Layers: **2**

4.2 Bahdanau Attention

Bahdanau Attention computes alignment scores between the decoder hidden state and all encoder outputs.

Instead of relying only on the encoder's final hidden state, attention enables the decoder to dynamically focus on the most relevant source words during translation.

4.3 Decoder

The decoder consists of:

- Embedding Layer
- LSTM
- Attention Context Vector
- Fully Connected Output Layer

Teacher forcing was applied during training with a ratio of 0.5 to improve convergence.

4.4 Beam Search

During inference, Beam Search with beam width **5** was implemented.

Beam Search keeps multiple candidate translations instead of selecting only the highest-probability word at each step, generally producing better translations than greedy decoding.

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5. Training Procedure

The model was trained using the following configuration.

Hyperparameter	Value
Embedding Size	256
Hidden Size	512
Encoder Layers	2
Decoder Layers	1
Batch Size	64
Maximum Sequence Length	50
Learning Rate	0.001
Optimizer	Adam
Dropout	0.3
Teacher Forcing Ratio	0.5
Epochs	20
Beam Width	5

Cross Entropy Loss was used as the optimization objective while ignoring padding tokens.

Gradient clipping with a maximum norm of 1.0 was applied to stabilize training.

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6. Experimental Results

Model Complexity

Trainable Parameters:

57,365,525

Training Performance

Final Training Loss:

2.0873

Final Validation Loss:

8.1910

The training loss consistently decreased over the training epochs, indicating that the model successfully learned patterns from the training data. However, the validation loss increased after several epochs, suggesting overfitting.

Evaluation Metrics

Metric	Score
BLEU Score	**0.0046**
BERTScore (F1)	**-0.1957**
Inference Time	**233.92 seconds**

Although the model completed training successfully, the BLEU and BERTScore indicate that translation quality remains limited.

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7. Translation Examples

Sanskrit Sentence	Reference Translation	Model Prediction
Ctrl, S नुत्वा रक्षन्तु।	Save it with Ctrl, S.	In this tutorial, I am using Ubuntu Linux OS...
गुरुः छात्रान् एकवारं पाठयति ।	Teacher will teach the students only once.	And when I say unto you...
वयं Colors विकल्पं तस्योपरि नोदनेन चिनुमः ।	I will choose Colors options by clicking on it.	This is the...

These examples demonstrate that the current model tends to generate repetitive high-frequency English phrases instead of accurate translations.

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8. Discussion

The implemented Seq2Seq model successfully integrates a Bidirectional LSTM encoder, Bahdanau Attention mechanism, and LSTM decoder into a complete neural machine translation system.

The decreasing training loss demonstrates that the network learned meaningful representations from the training data. However, the increasing validation loss and low BLEU score suggest that the model overfitted the training data and did not generalize well to unseen examples.

Several factors may have contributed to the limited translation performance:

- Small parallel corpus.
- Simple whitespace tokenization.
- Large vocabulary leading to data sparsity.
- Rich morphology of Sanskrit.
- Limited training duration relative to model size.

Despite these limitations, the project successfully demonstrates the implementation of a complete attention-based neural machine translation pipeline.

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9. Future Improvements

Translation quality can be improved through:

- Better tokenization using SentencePiece or Byte Pair Encoding (BPE).
- Vocabulary pruning.
- Early stopping based on validation BLEU.
- Learning rate scheduling.
- Regularization techniques.
- Larger Sanskrit-English parallel corpora.
- Transformer-based neural machine translation models.
- Pretrained multilingual language models.

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10. Conclusion

This project presented a complete implementation of a Sanskrit-to-English Neural Machine Translation system using a custom Seq2Seq architecture with Bahdanau Attention. The model was trained and evaluated using the provided dataset and successfully completed the end-to-end translation pipeline, including preprocessing, training, beam search decoding, and evaluation.

Although the current translation accuracy is limited, the implementation satisfies the objectives of building a custom neural machine translation system and provides a strong foundation for future improvements using advanced tokenization techniques, larger datasets, and modern transformer architectures.

References

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